CS182: Introduction to Machine Learning
Lecture 06: Parameter Estimation

Lecture Quiz and Answer Copy

Scope: Parameter_Estimation.pdf. Independent practice material based on the official 2026 Fall lecture.

Total: 100 marks. **Suggested time:** 120 minutes.

Name: _____ **Student ID:** _____

Section	Questions	Each	Marks
A. Misconceptions: True/False	12	1	12
B. Multiple response	12	2	24
C. Proofs and formula derivations	4	10	40
D. Applications	2	12	24
Total	30		100

Scoring. In Section B each item has exactly four options, of which one, two, or three are correct. The complete correct set earns 2 marks; a nonempty proper subset earns 1 mark; any incorrect option or no selection earns 0. T/F answers earn 1 mark only when correct. Show reasoning in Sections C and D; part marks are available for valid intermediate work. Equivalent correct methods receive credit.

Instructions. Answer in English. Use natural logarithms. Exact values and justified approximations are accepted. All vectors are columns unless stated otherwise. Use local assumptions, label conventions, and tie rules.

Answer sheet — Section A

1	2	3	4	5	6

7	8	9	10	11	12

Answer sheet — Section B

13	14	15	16	17	18

19	20	21	22	23	24

Section A. True/False 12 marks

1. A full $d \times d$ empirical covariance estimated within a class from exactly d observations is necessarily invertible if all feature sample variances are positive.

2. The posterior mean minimizes posterior expected loss for every possible loss function.

3. Zero covariance between two variables guarantees their independence without any assumption about their joint distribution.

4. With positive class priors and a shared positive-definite covariance, Gaussian log-discriminant differences are affine, even when the class priors are unequal.

5. If the posterior for an unknown Gaussian mean is normal, substituting its mean into the likelihood reproduces the full Bayesian posterior predictive variance.

6. Filling missing values of one feature with its observed sample mean leaves that feature's maximum-likelihood sample variance unchanged.

7. An unbiased estimator necessarily has smaller mean squared error than every biased estimator of the same parameter.

8. For an unrestricted categorical model on a closed probability simplex, a category observed zero times has maximum-likelihood probability zero.

9. For observed data D, the likelihood $L(\theta; D)$ is generally a probability density over θ and must integrate to 1 with respect to θ.

10. If every Gaussian class has a diagonal covariance matrix, the classifier must have linear decision boundaries even when the diagonal variances differ across classes.

11. When the Gaussian mean is estimated from the same N observations, the maximum-likelihood variance with denominator N is unbiased for the population variance.

12. A regression model using fixed features $1, x, x^2, \sin x$ can still be linear in its unknown parameters.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

13. A categorical variable has counts $(3, 0, 7)$ in a sample of size 10. The parameter space is the closed simplex. Select every correct statement.

(A) At the MLE, the zero-count term in the log-likelihood is interpreted as zero.

(B) The MLE is $(0.3, 0, 0.7)$.

(C) The simplex constraint can be discarded because the category likelihoods are independent functions of the parameters.

(D) A Lagrange-multiplier calculation that assumes every $p_i > 0$ needs a separate boundary argument for the unseen category.

14. Select every correct statement about the regression likelihood and design matrix.

(A) Independent Gaussian errors with known unequal variances lead to weighting each squared residual by the inverse of its variance.

(B) In a joint likelihood, an input-density term $p(x \mid \theta)$ may always be dropped when fitting θ.

(C) Independent Gaussian errors with one fixed, known common positive variance make maximum likelihood in the mean parameters equivalent to ordinary least squares.

(D) Fixed nonlinear basis functions of the inputs can be used while retaining a model linear in its coefficients.

15. A nonconstant Gaussian sample has $N = 5$, sample mean $m = 2$, and $\sum_t (x^t - m)^2 = 20$. Both the mean and variance are unknown. Select every correct statement.

(A) The usual unbiased variance estimate is 5.

(B) The denominator $N - 1$ is required to maximize the Gaussian likelihood.

(C) The maximum-likelihood standard deviation is $\sqrt{5}$.

(D) The variance MLE is 4.

16. Assume a bivariate Gaussian with positive-definite covariance. Select every correct statement.

(A) For $\Sigma = \mathrm{diag}(9, 1)$, the squared Mahalanobis distance from μ to $\mu + (3, 1)^\mathsf{T}$ is 2.

(B) A zero off-diagonal covariance makes these two jointly Gaussian coordinates independent.

(C) For the same covariance and displacement, the squared Euclidean distance is also 2.

(D) A positive correlation tilts the long axis of the Gaussian contour ellipse toward positive slope.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

17. A Gaussian mean has prior $\theta \sim N(1, 4)$. Three conditionally independent observations have distribution $N(\theta, 9)$ and sample mean 4. Select every correct statement.

(A) The posterior variance is $12/7$.

(B) The posterior mean is $19/7$.

(C) The posterior MAP estimate equals the sample mean 4.

(D) The full posterior predictive variance of a new observation is 9.

18. For two independent Bernoulli features conditional on class, use priors $0.6, 0.4$, class C_1 feature probabilities $(0.5, 0.8)$, and class C_2 probabilities $(0.9, 0.3)$. Observe $x = (0, 1)$. Select every correct statement.

(A) For probabilities strictly between 0 and 1, the log discriminant is affine in the binary coordinates.

(B) The zero-valued first feature contributes no likelihood factor.

(C) The joint class-feature scores are 0.24 and 0.012.

(D) The posterior probability of C_1 is 0.24.

19. Select every correct statement about model selection.

(A) The model evidence is generally just the largest likelihood over that model's parameters.

(B) A validation fold should be included in fitting the very predictor whose error it is intended to assess.

(C) In a regularized objective, the penalty weight controls the strength of the preference for the penalized notion of simplicity.

(D) Bayesian model posterior probabilities are proportional to model evidence times model prior probabilities.

20. Five i.i.d. Bernoulli observations are all equal to 1. Select every correct statement about maximum likelihood.

(A) The absence of an interior stationary point proves that no MLE exists on the closed interval.

(B) On the closed parameter interval $0 \le p \le 1$, the MLE is $p = 1$.

(C) The log-likelihood has a stationary point in $0 < p < 1$.

(D) On the open parameter interval $0 < p < 1$, the maximum occurs at $p = 1 - 1/N = 4/5$.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

21. A Gaussian classifier has $K = 3$ classes and $d = 4$ features. Count independent covariance parameters only, unless stated otherwise. Select every correct statement.

(A) The three class priors add three independent parameters.

(B) One shared spherical covariance $s^2 I$ requires 4 covariance parameters.

(C) Separate full covariance matrices require 30 parameters.

(D) One shared full covariance matrix requires 16 independent covariance parameters.

22. At a fixed test input, assume a new test response is independent of the training set and has conditional mean $f(x)$ and variance $\sigma^2(x)$. Select every correct statement.

(A) Increasing model complexity is a theorem that strictly increases estimator variance for every learning algorithm and distribution.

(B) The expected test error is noise variance plus squared prediction bias plus the variance over training sets of the predictor.

(C) If every trained predictor equals $f(x)$, the expected squared test error still equals $\sigma^2(x)$.

(D) The smallest error on a finite validation set identifies the population-optimal model with certainty.

23. For $N = 4$ i.i.d. normal observations with true variance 1, let $Q = \sum_t (X_t - \bar{X})^2$. You may use $E[Q] = 3$ and $\mathrm{Var}(Q) = 6$. Compare $V_{\mathrm{ML}} = Q/4$ and $V_{\mathrm{U}} = Q/3$. Select every correct statement.

(A) This example shows that the unbiased estimate must have the smaller MSE.

(B) Under i.i.d. sampling with finite variance, the sample mean is consistent even though an arbitrary single observation need not be close to the mean.

(C) The MSE of V_{ML} is $7/16$.

(D) The MSE of V_{U} is $2/3$.

24. There are $d = 5$ features, $K = 3$ classes, and four observations in each class. Covariances are computed by centering within class. Select every correct statement.

(A) Increasing a class to six observations is by itself sufficient to guarantee a nonsingular full covariance.

(B) The shared pooled full covariance must also be singular.

(C) Every separate full class covariance estimate is singular.

(D) A diagonal covariance approximation can be invertible even when every separate full covariance is singular.

Section C. Proofs and Formula Derivations

25. Discrete likelihoods, constraints, and naive Bayes **[10 marks]**

(a) For $N \geq 1$ i.i.d. Bernoulli observations, let $s = \sum_t x^t$. Derive the MLE on the closed interval $[0, 1]$, treating $0 < s < N$, $s = 0$, and $s = N$ separately. Establish that your candidates are global maxima. **[3 marks]**

(b) For a categorical distribution with counts $n_1, \ldots, n_K$, $\sum_k n_k = N > 0$, derive the constrained MLE. Include a justification for categories having zero counts rather than assuming an interior solution. **[4 marks]**

(c) For binary features conditionally independent given class, derive the class-conditional likelihood, the count-based class-specific MLEs, and an affine log discriminant in the binary input coordinates. State the condition for finite log coefficients. **[3 marks]**

Student response

(a)

(b)

(c)

Section C. Proofs and Formula Derivations

26. Gaussian variance and two kinds of bias-variance decomposition [10 marks]

(a) Let $X_1, \ldots, X_N$ be i.i.d. $N(\mu, v)$, with unknown μ, $v > 0$, and $N \geq 2$. Use $\sum_t (X_t - \mu)^2 = \sum_t (X_t - \bar{X})^2 + N(\bar{X} - \mu)^2$ to profile out μ, then derive the variance MLE when $T = \sum_t (X_t - \bar{X})^2 > 0$. What happens if $T = 0$? **[3 marks]**

(b) Assuming only i.i.d. observations with mean μ and finite variance v, prove $E[T] = (N-1)v$. Deduce the bias of T/N and an unbiased correction. **[2 marks]**

(c) For a scalar estimator $d(D)$ of a fixed parameter θ, derive $\mathrm{MSE}(d) = \mathrm{Var}(d) + [E(d) - \theta]^2$, and explain why unbiasedness alone does not rank estimators by MSE. **[2 marks]**

(d) At a fixed input x, let an independent future response Y have conditional mean $f(x)$ and variance $\sigma^2(x)$. A predictor $g_D(x)$ is fitted to a random training set D. Derive the expected squared test-error decomposition over both Y and D. Distinguish its bias from parameter-estimator bias.**[3 marks]**

Student response

(a)

(b)

(c)

(d)

Section C. Proofs and Formula Derivations

27. A Gaussian prior is more than a point estimate **[10 marks]**

(a) Suppose $X_t \mid \theta$ are conditionally i.i.d. $N(\theta, \sigma^2)$, with known $\sigma^2 > 0$, and $\theta \sim N(m_0, \tau^2)$, $\tau^2 > 0$. Complete the square in the likelihood times prior to derive the posterior mean and variance, including their precision form. **[4 marks]**

(b) For a new conditionally independent observation $X_* \mid \theta \sim N(\theta, \sigma^2)$, derive the posterior predictive distribution. Compare its variance with the distribution obtained by substituting the posterior mean into the observation model. **[3 marks]**

(c) Derive the minimum-posterior-MSE point estimator of θ, identify its relationship to MAP in this model, and find the limits of m_N, v_N as the prior becomes diffuse with N fixed. Does that limit make the finite-sample plug-in predictive distribution exact? **[3 marks]**

Student response

(a)

(b)

(c)

Section C. Proofs and Formula Derivations

28. Gaussian discriminants and when covariance estimates can be used [10 marks]

(a) For Gaussian class models $N_d(m_i, S_i)$ with positive-definite S_i and positive class priors π_i, derive the quadratic log discriminant from the density. Express it as $x^\mathsf{T} W_i x + w_i^\mathsf{T} x + b_i$. **[3 marks]**

(b) Assume all classes share S. Derive the affine LDA scores, the pairwise boundary equation, and the nearest-mean interpretation for equal priors. State when this becomes ordinary Euclidean nearest mean. **[3 marks]**

(c) Let all K classes be represented, with N_i observations each and total N. Count free covariance parameters for separate full versus shared full covariances. Then prove the rank bounds for a class covariance and for $S = \sum_i (N_i/N) S_i$. Give necessary sample-size conditions for invertibility and explain why they are not sufficient. **[4 marks]**

Student response

(a)

(b)

(c)

Section D. Applications

29. Estimating a Gaussian classifier from rectangular samples **[12 marks]**

(a) Class C_1 contains the four observations $(-1,-2),(-1,2),(1,-2),(1,2)$. Class C_2 contains $(1,0),(1,4),(3,0),(3,4)$. Fit maximum-likelihood class priors, means, separate covariance matrices, and the pooled covariance used by LDA. Use the likelihood denominator rather than an unbiased denominator. **[4 marks]**

(b) Derive a simplified expression for $g_2(x) - g_1(x)$ and the decision boundary. State which side is assigned to C_2, and explain whether fitting QDA to these particular estimates changes the classifier. **[3 marks]**

(c) Classify $x_* = (0.5, 2)^\mathsf{T}$. Compare its squared Mahalanobis distances and squared Euclidean distances to the two means, and explain any disagreement between the resulting nearest-mean decisions. **[2 marks]**

(d) For two classes in two dimensions, count covariance parameters in separate full, shared full, shared diagonal, and shared spherical models. Which of these covariance assumptions exactly matches the estimates above? Check the necessary class and pooled sample-size inequalities, and say whether those inequalities alone establish invertibility. **[3 marks]**

Student response

(a)

(b)

(c)

(d)

Section D. Applications

30. Gaussian regression, prediction error, and model selection **[12 marks]**

(a) A regression model is $Y = w_0 + w_1 x + \varepsilon$, with independent $\varepsilon \sim N(0, \sigma^2)$, $\sigma^2 > 0$, and an input distribution independent of the fitted parameters. Starting from the joint likelihood, derive the least-squares objective and its normal equations for the coefficient vector $w = (w_0, w_1)^\mathsf{T}$.**[3 marks]**

(b) For observations $(x, y) = (-1, 0), (0, 1), (1, 2), (2, 5)$, compute the fitted line, its residual sum of squares, and the Gaussian observation-variance MLE when variance is estimated jointly with the line. **[3 marks]**

(c) In a separate repeated-training study at a fixed input, the true conditional mean is 2 and test-noise variance is 0.25. Degree-1, degree-2, and degree-3 polynomial learners have training-set-averaged predictions $1.7, 1.9, 2.0$, and prediction variances $0.09, 0.12, 0.50$, respectively. Compute each expected squared test error and choose the best learner at this input. **[3 marks]**

(d) Three equally sized validation folds give mean squared errors $(1.0, 1.4, 1.2)$ for degree 1, $(0.6, 0.9, 0.9)$ for degree 2, and $(0.4, 1.5, 1.7)$ for degree 3. Choose a degree by cross-validation, describe the fitting/validation separation required, and explain why a smaller training error for degree 3 would not settle the choice. **[3 marks]**

Student response

(a)

(b)

(c)

(d)